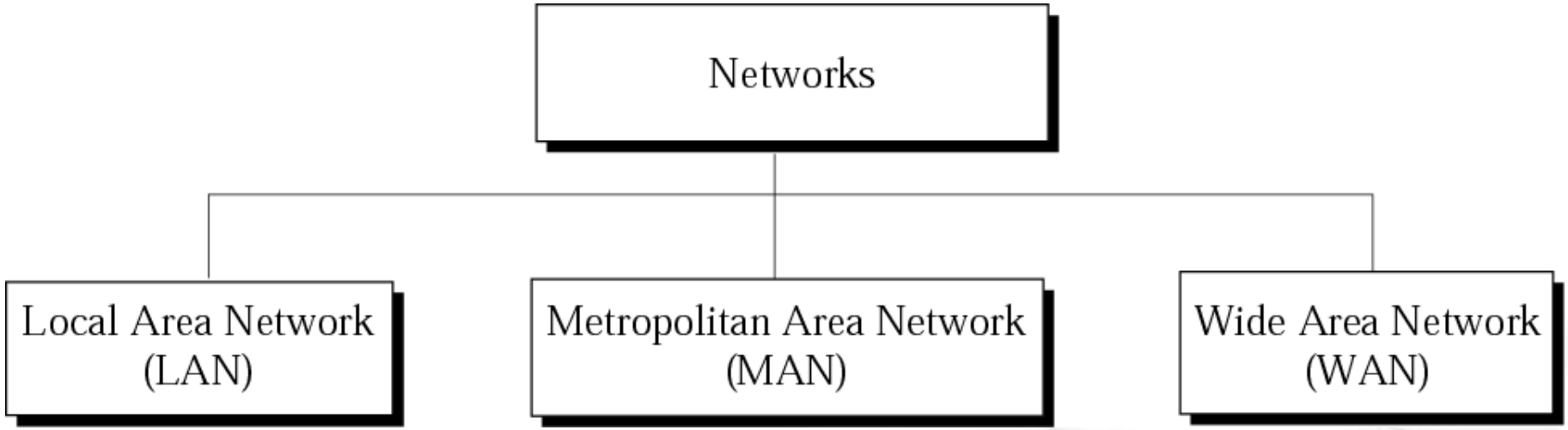
Network Security
Network Architecture

DaeHun Nyang
Ewha WU.

Networks

Classification of Networks



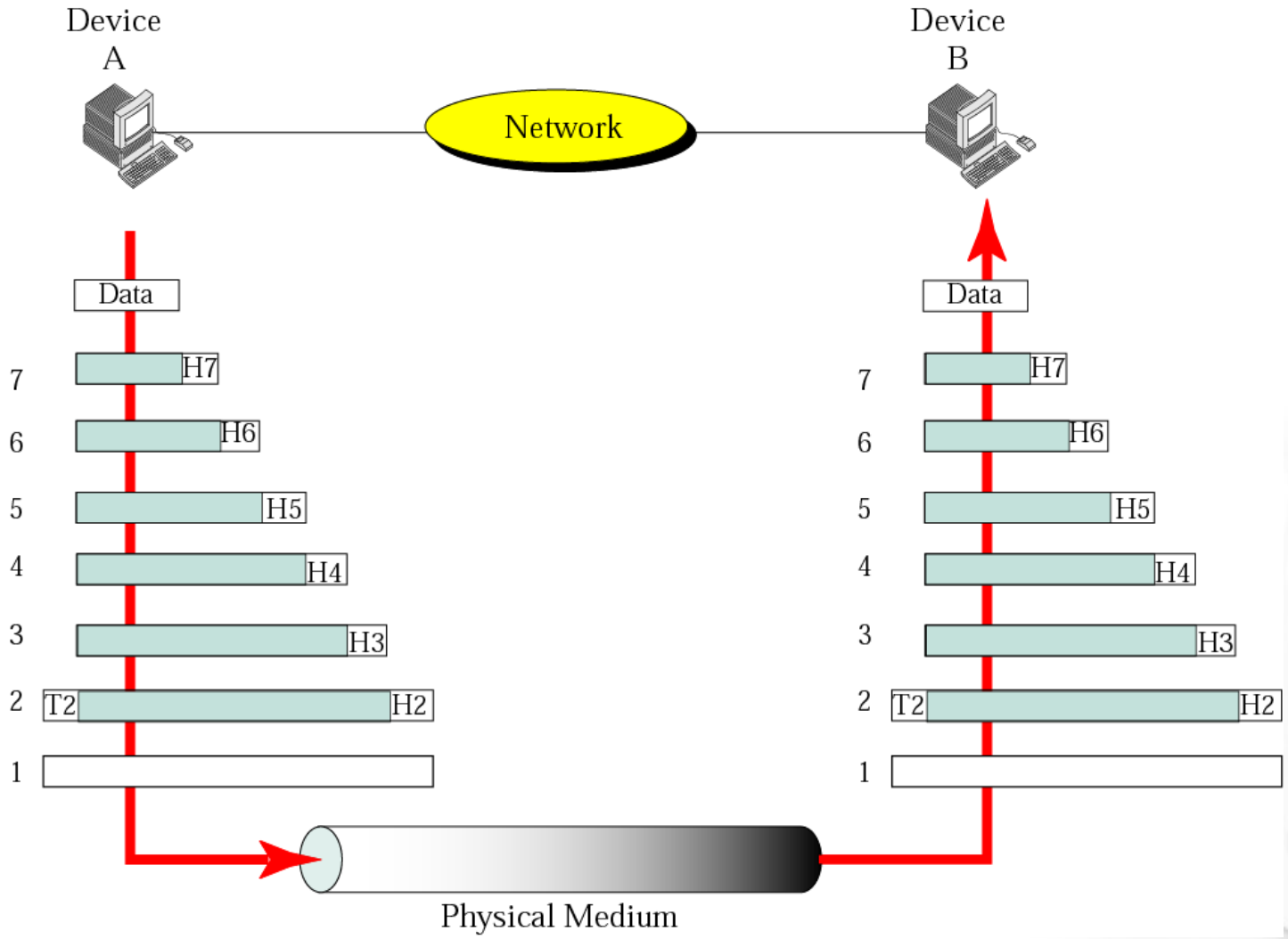
OSI Reference Model

Seven layers

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

OSI Reference Model

Data flow in OSI Model



OSI Model

PHY Layer

- Bits are grouped together to form a bit stream and converted into electrical signals that can be transmitted on a physical medium
- Determined the mechanical or physical specifications for network hardware

예 10BASE-T, 1000BASE-T, RS-232 (serial), …

10BASE-T cable and jack
(twisted pair)

OSI Model

L2 (Data Link)

- Bits are organized into units called frames, which can be used to transfer data directly between neighboring computers. (e.g., Ethernet protocol).
- Using a physical address or MAC address

L3 (Network)

- Responsible for sending between a specific destination and origin
- Each computer is assigned information called a logical address or IP address.
- Data is sent in logical units called packets.

L4 (Transport)

- Responsible for ensuring messages get from source to destination (source-to-destination delivery)
- Split large messages into multiple packets for transmission
- Reorder packets if they arrive out of order
- Request retransmission of packets if they are lost

OSI Model

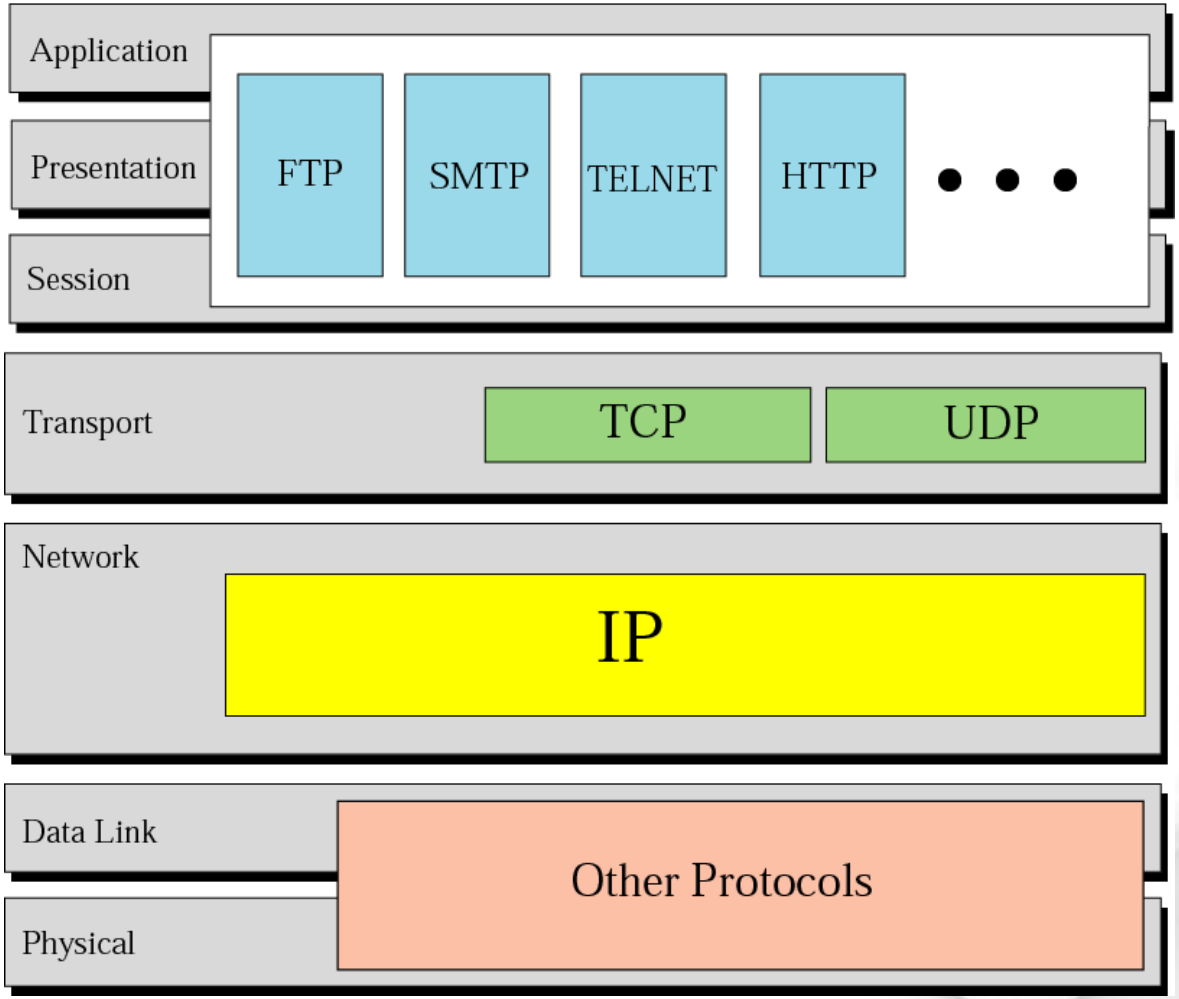
L5,6 (Session and Presentation)

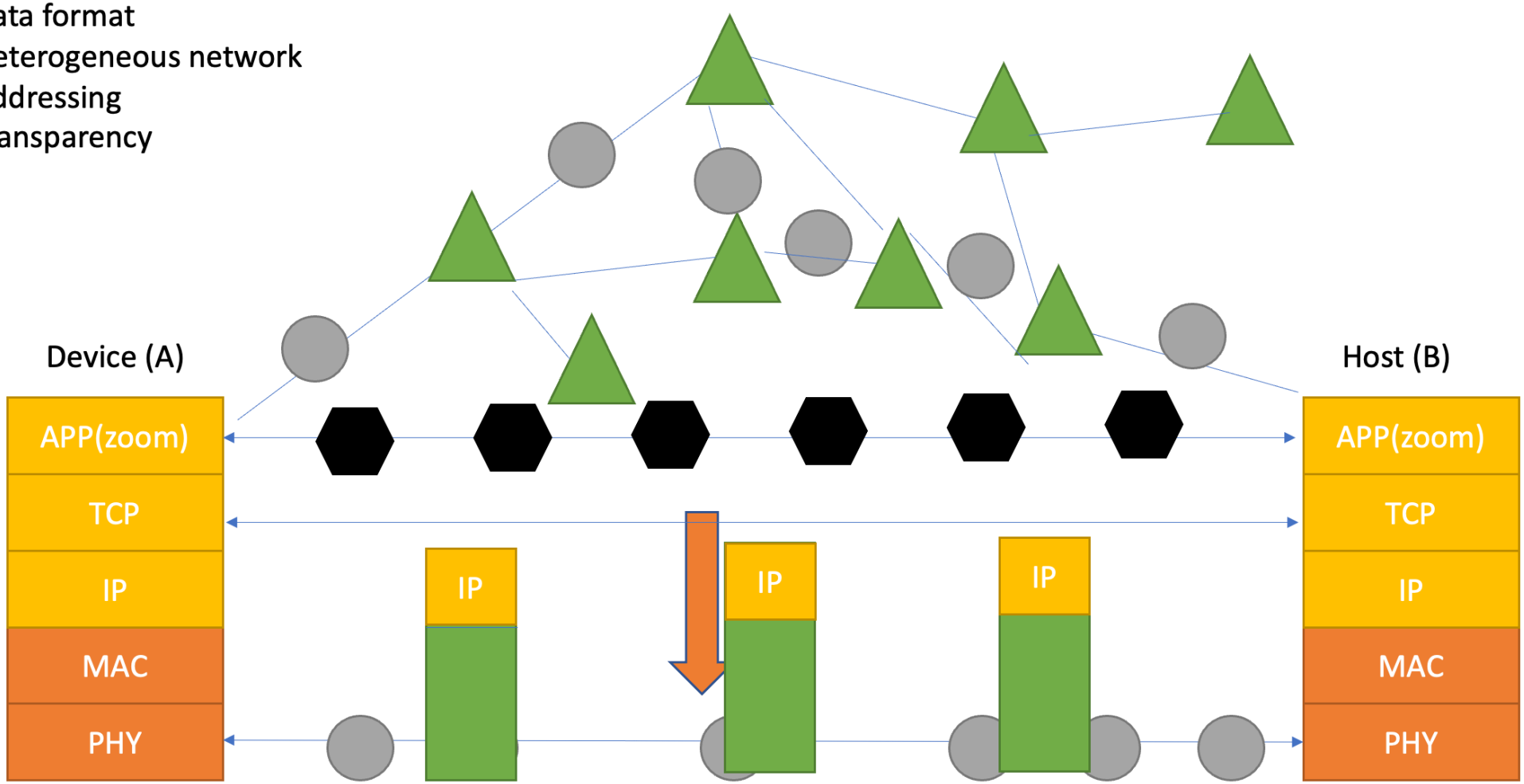
- In today's networks, most of them are not implemented, other layers are taking over the functions of these layers.

L7 (Application Layer)

- Allow users or application software to use the network.
  - File transfer (FTP)
  - e-mail (SMTP)
  - Web (HTTP)

TCP/IP and OSI model



Security issue
Routing issue
Packet loss
Data format
Heterogeneous network
Addressing
Transparency

Five tuple – (srcIP, dstIP, pType, srcPort, dstPor)